Assignment specification

Exam type:

The exam should be handed in two parts: Implementation (.ipynb code); and a report (pdf) of max 12 pages (on both parts). The provided code should be an executable solution; the structure and the content of the report should follow as acceptable academic style and should cover all the required tasks.

You are expected to draw upon relevant readings from the course. Other readings and references are also allowed.

Supervised Learning Exam part (50 marks)

1. Task One: (Code + Report)

Based on your Dataest (previously chosen), visualize it, perform feature extraction and representation, and choose a model to perform supervised prediction/classification, the implementation should follow the following steps (provide the implementation as .jpynb):

• Load the Dataset file, access the columns and convert them into proper data types in Python, whenever needed.
• Perform the necessary pre-processing.
• Apply a selected model on the data to perform the task.
• Evaluate the performance your model.

Then write a report explaining the following.

• Explain your data and features. Illustrate each step with proper visualization figures.
• Explain your model, the reason behind the model selection, explain why it suits the task (prediction or classification).
• Explain how your model works as well as its underlying concept.• Illustrate your results using tables/figures and discuss the model performance in terms of the selected metrics. Please explain why those metrics are effective to test the model performance.

2. Task Two: (Report)

From an ethical perspective and based and your above accomplished analysis:

• What are the required measures to ensure that your analysis complies with the AI ethical considerations, please discuss this for each phase of your pipeline and reflect.
• How could you improve your pipeline into a more sustainable machine learning framework? Please explain in clear steps what should be modified.

Semi- and Unsupervised Learning Exam (50 marks)

1. Semi-Supervised Task (Code + report). 10 marks

Consider an image classification problem with 1,000 training images. Only 200 images have class labels, while the remaining 800 images are unlabelled.

• Basic understanding of semi-supervised learning

Present your understanding of semi-supervised learning. Explain, in simple terms, how semi-supervised learning is different from supervised learning and unsupervised learning.

• Pseudo-labelling workflow

Explain the basic steps of pseudo-labelling as a simple semi-supervised learning method. Your explanation should describe how the labelled data and unlabelled data are used during training. You may use a simple diagram to illustrate the workflow.

• Advantage and limitation

Explain one advantage and one limitation of using pseudo-labelling. In particular, explain what may happen if the model assigns an incorrect pseudo-label to an unlabelled sample.

• Simple programming task

Use the built-in Iris dataset from scikit-learn and perform pseudo-labelling experiment. Provide the implementation as an executable Jupyter Notebook (.ipynb). Use the following steps:

▪ Load the Iris dataset and split it into a training set and a test set.

▪ Use only 30% of the training samples as labelled data. Treat the remaining 70% of the training samples as unlabelled.
▪ Train a Random Forest classifier using only the labelled training samples and report its test accuracy.
▪ Use the trained classifier to predict the unlabelled samples. Keep only predictions with confidence of at least 0.90 and use them as pseudo-labels.
▪ Add the accepted pseudo-labelled samples to the labelled training set, retrain the same classifier, and report the new test accuracy.
▪ Briefly compare the accuracy before and after pseudo-labelling and state whether pseudo-labelling helped in your experiment.
Expected submission2. ▪ A short report answering Tasks a-c and briefly discussing the programming result in Task d.
▪ An executable Jupyter Notebook (.ipynb) containing the code for Task d.
▪ Simple tables, plots, training curves and figures may be included where useful.
Unsupervised Task (Code + Report). 40 marks
Question 2.1:
▪ Explain, in simple terms, what K-means does: state the objective it minimizes and the two alternating steps of the algorithm. Explain what makes clustering unsupervised, and how the question it answers differs from the question a classifier answers on the same data.
▪ Explain what factors affect the quality of the K-Means clustering final result. And describe how to determine what is the best K to use in your analysis.
▪ Cluster Wine Dataset:
• Load the Wine dataset (13 chemical features, 178 wines). Standardize the features.
• Run K-means for k = 2 … 8. State which k you select and justify the choice using analysis in plots.
• For your selected k, report the cluster sizes and visualize the clusters in a 2-D scatter of two features of your choice.
• External evaluation only: build a contingency table of cluster label vs. cultivar and report the Adjusted Rand Index (ARI). Comment on how well the clusters recover the cultivars.
Question 2.2:
▪ Explain what is PCA and what it computes. Describe the algorithm conceptually and its use-case. How .
▪ what “explained variance” of a component means. Explain why features are standardized before PCA .
▪ What are the assumptions PCA needs to work properly?
▪ How many principle components PCA produces and what is your criterium to determine how many are you going to keep in your analysis?
▪ PCA Analysis using Breast cancer dataset:
• Standardize the 30 features and fit a full PCA.
• Plot the scree curve (variance per component) and the cumulative explained variance. Report the variance of PC1 and PC2 and the number of components needed for a proper analysis.
• Plot the data projected onto PC1–PC2, colored by diagnosis. Describe what you see.
• Report the five features with the largest absolute loadings on PC1 and give PC1 a short, meaningful name based on them.

Grade	Learning Outcome 1: Knowledge	Learning Outcome 2: Skills	Learning Outcome 3: Competence
A Excellent	Excellent and comprehensive	Demonstrates excellent analytical, technical and	Outstanding degree of judgment and independent

	understanding of concepts	writing skills	critical thinking
B Very good	Very good understanding of concepts	Demonstrates very good analytical, technical and writing skills	Sound degree of judgment and independent critical thinking
C Good	Good understanding of theory in most important areas	Demonstrates good analytical, technical and writing skills	Reasonable degree of judgment and independent critical thinking
D Satisfactory	Satisfactory understanding of theory, but with significant shortcomings	Demonstrates limited analytical, technical and writing skills	Limited degree of judgment and independent critical thinking
E Sufficient	Meets the minimum understanding of concepts	Demonstrates sufficient analytical, technical and writing skills	Very limited degree of judgment and independent critical thinking
F Fail	Fail to meet the minimum academic criteria.	No demonstration of analytical, technical and writing skills	Absence of judgment and independent critical thinking

Assignment criteria*

*Adapted from The Norwegian Association of Higher Education Institutions (http://www.uhr.no/utdanning/karakterpanel_1)

The assignment is worth 100 % of the grade of the course.
A report is requested as the submission, where the solutions covering both theoretical and practical tasks are included. Implementation should also be provided as appendix in the submitted document.
Marking Criteria: The below described weights are equally divided over the supervised and unsupervised part

1. Task One (70%): Code + Report
● Code quality and clarity (30%)
• Effective presentation of data, features, results (tables, graphs, visualizations) whenever applies (20%)
• Clearly defined methodology (10%)
• Results must be meaningfully compared and systematically analyzed (10%)
2. Task Two (30%): short essay
● Understanding theory and underlying principles (15%)
● Structured discussion (15%)